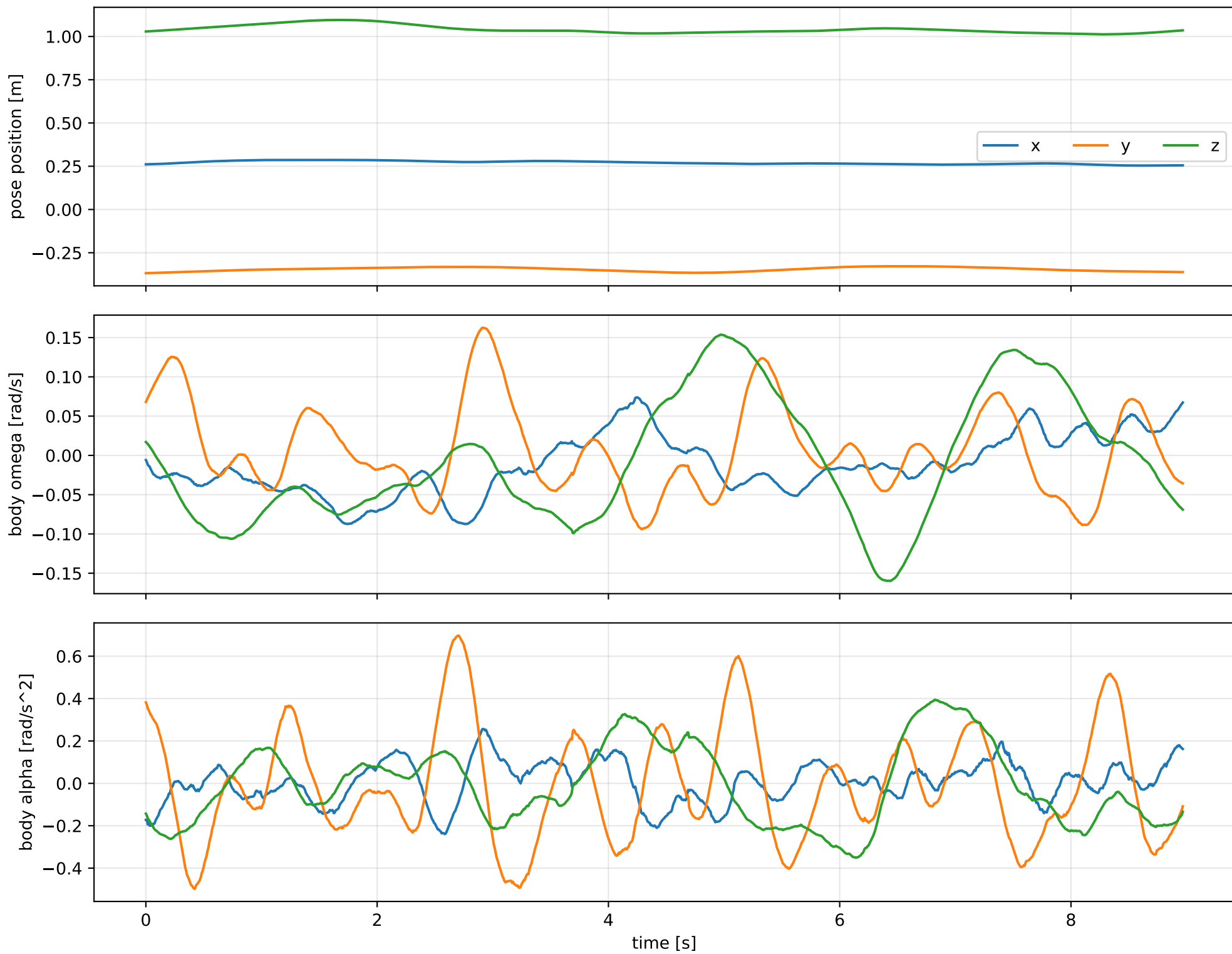
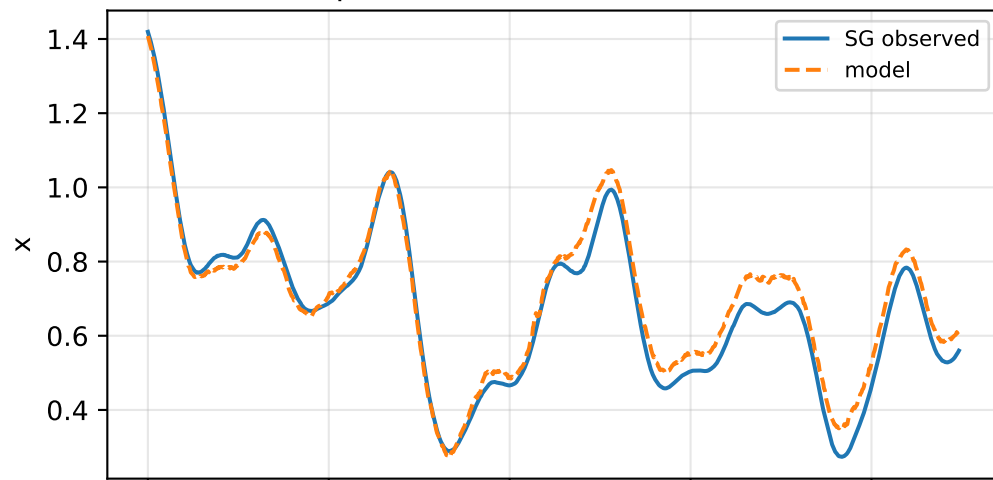


lag_fixed_one_period: SG trajectory and derived motion

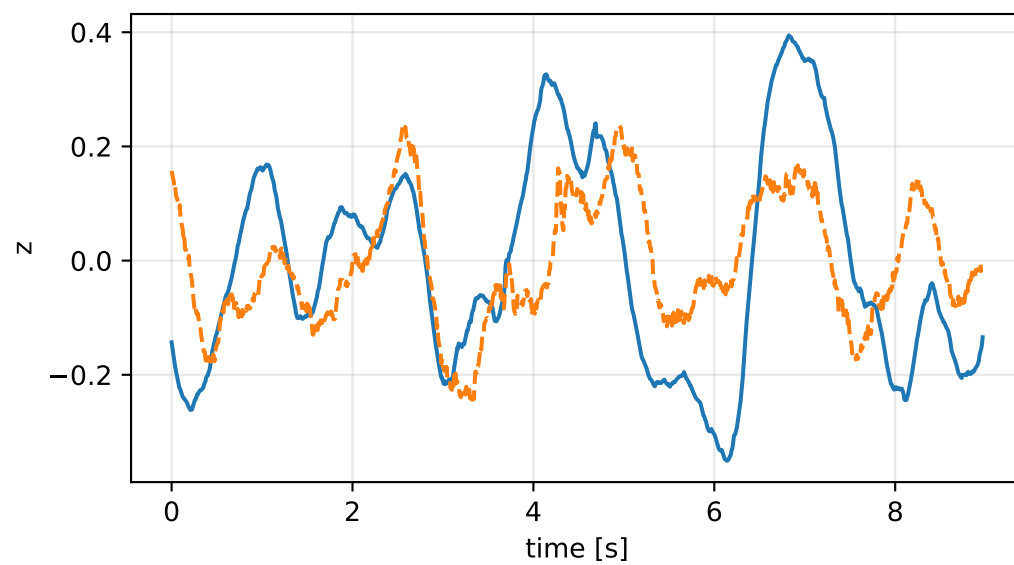
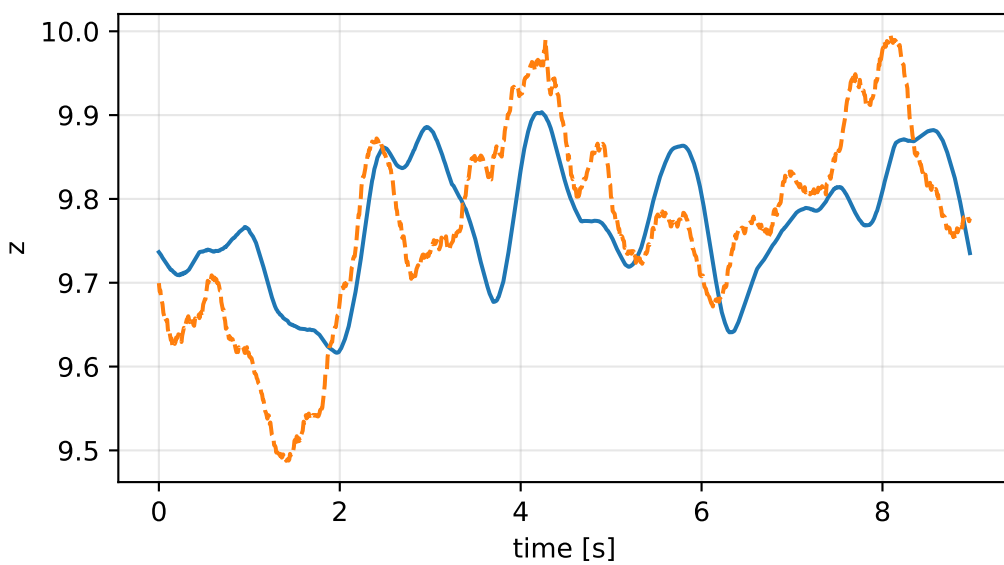
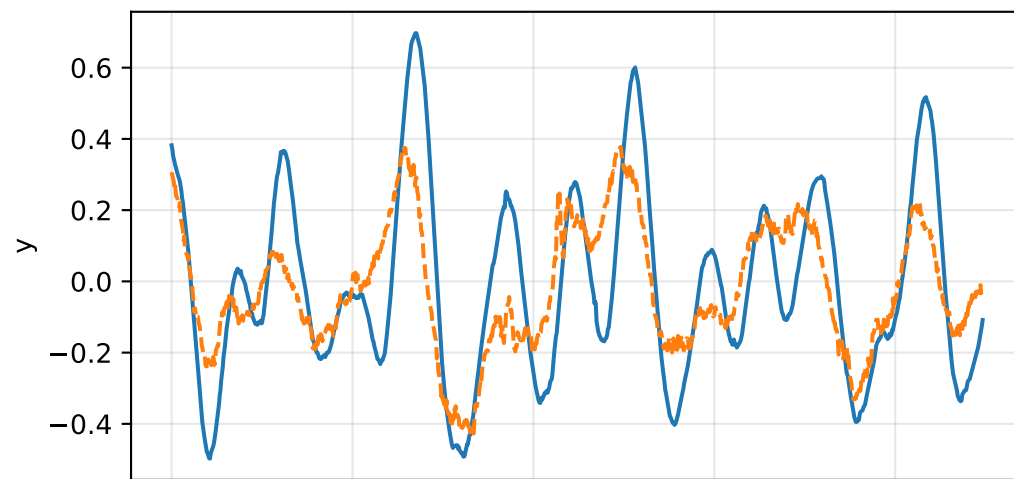
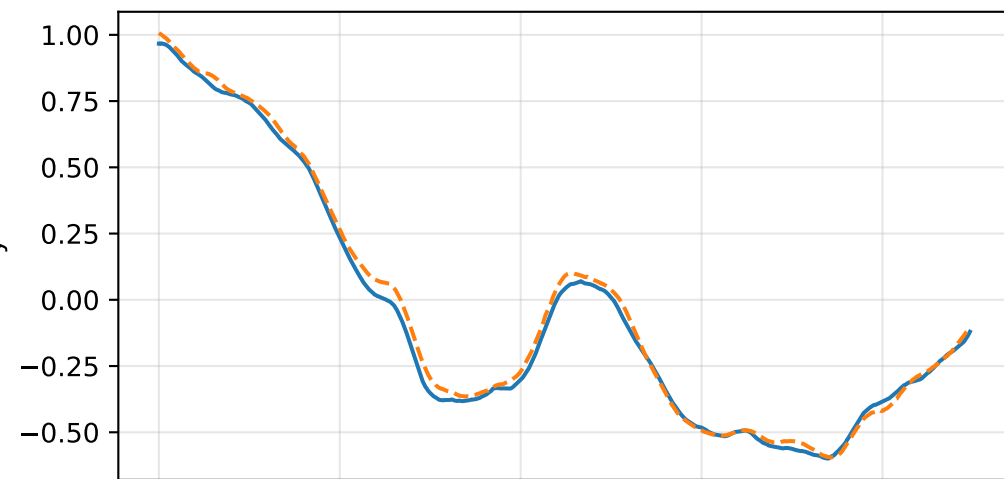
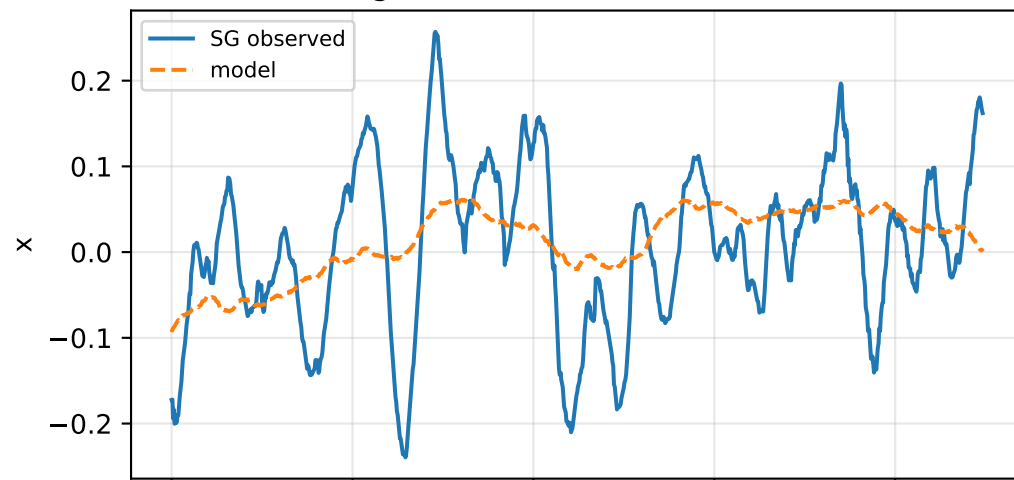


lag_fixed_one_period: acceleration residual objective

specific acceleration [m/s^2]

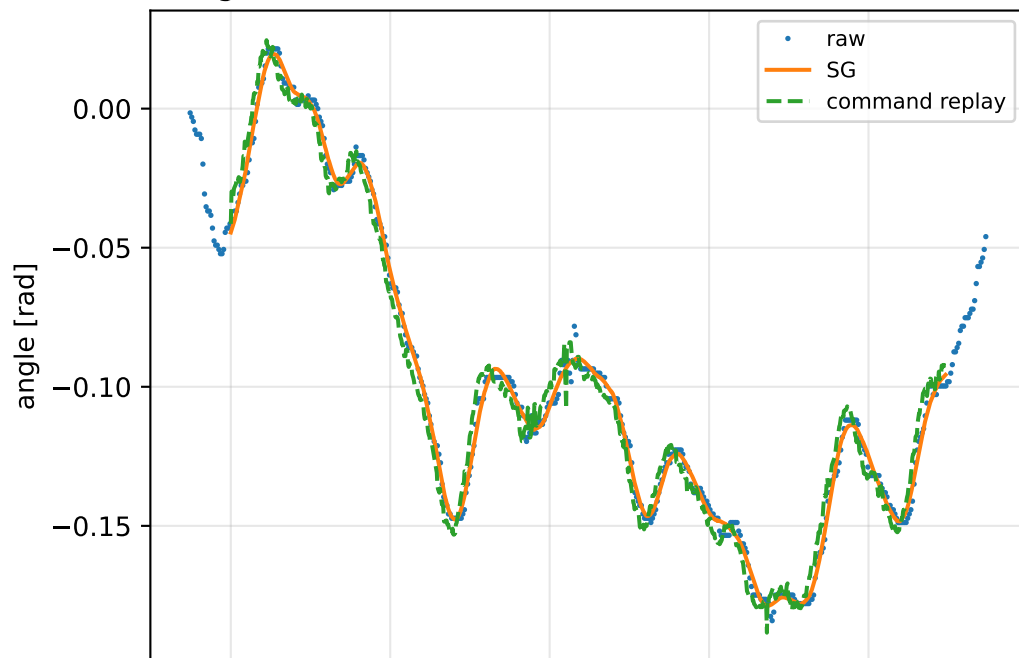


angular acceleration [rad/s^2]

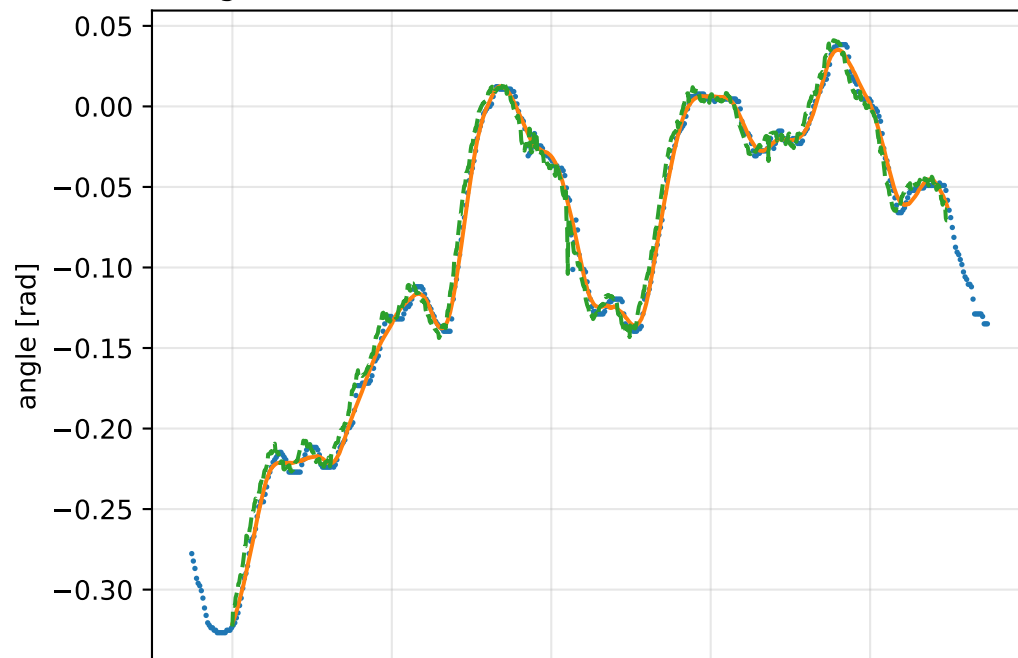


lag_fixed_one_period: gimbal measurement audit (objective=measured_sg)

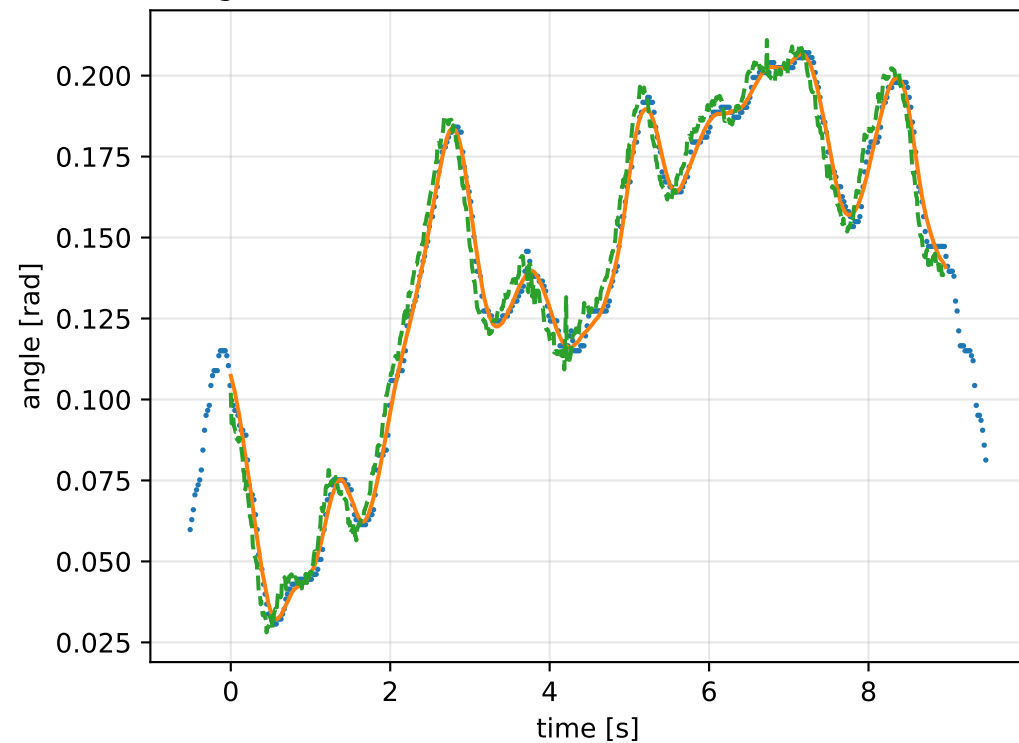
gimbal 1: RMSE=0.006345, max=0.01494 rad



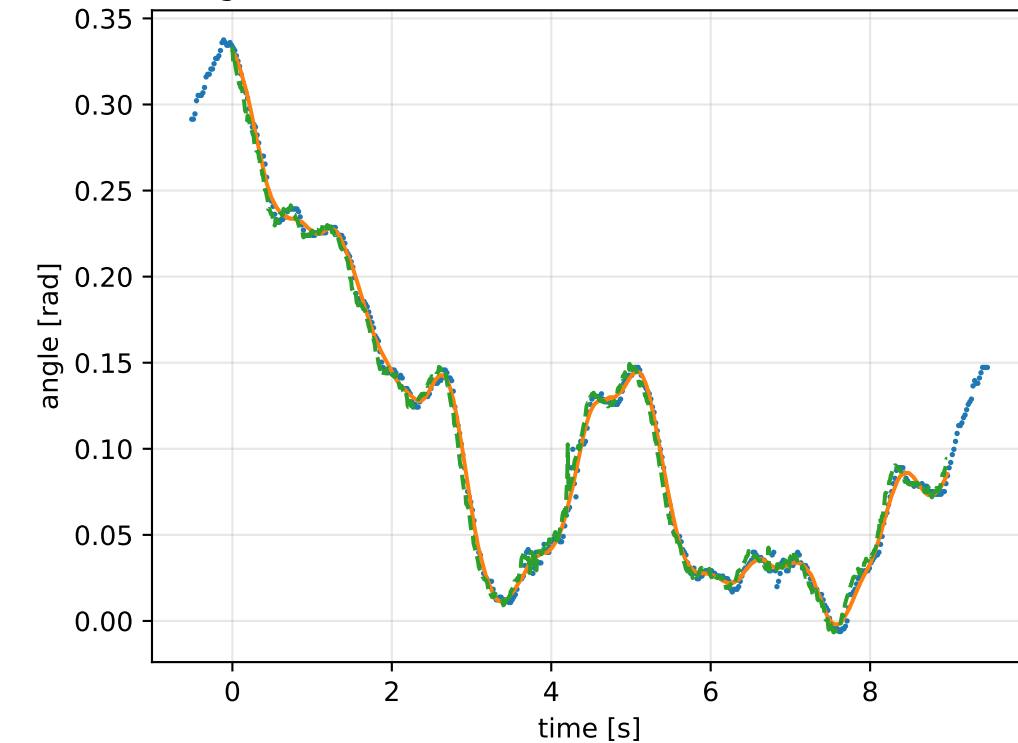
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

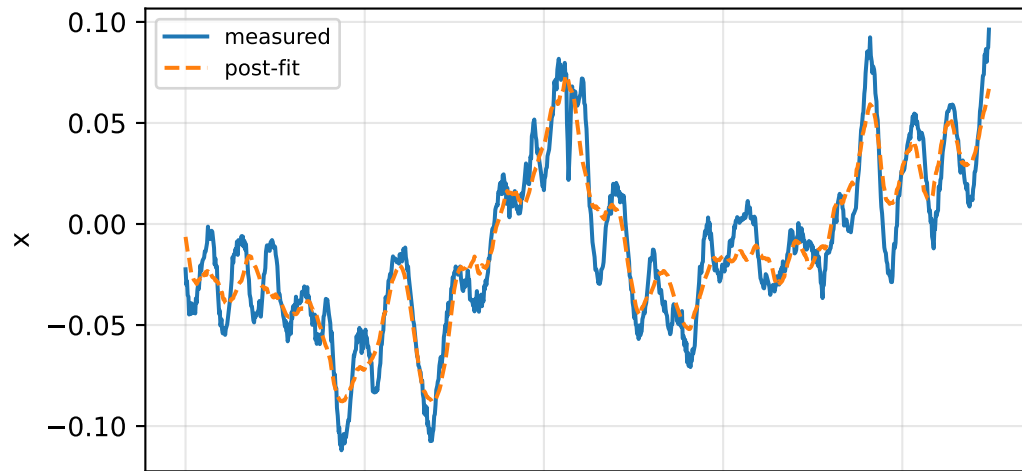


gimbal 4: RMSE=0.006321, max=0.03642 rad

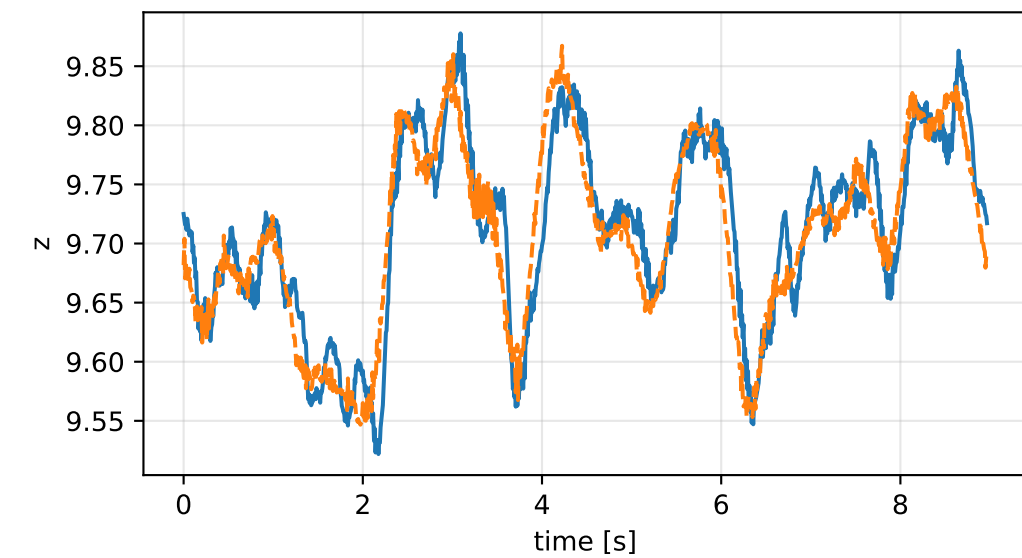
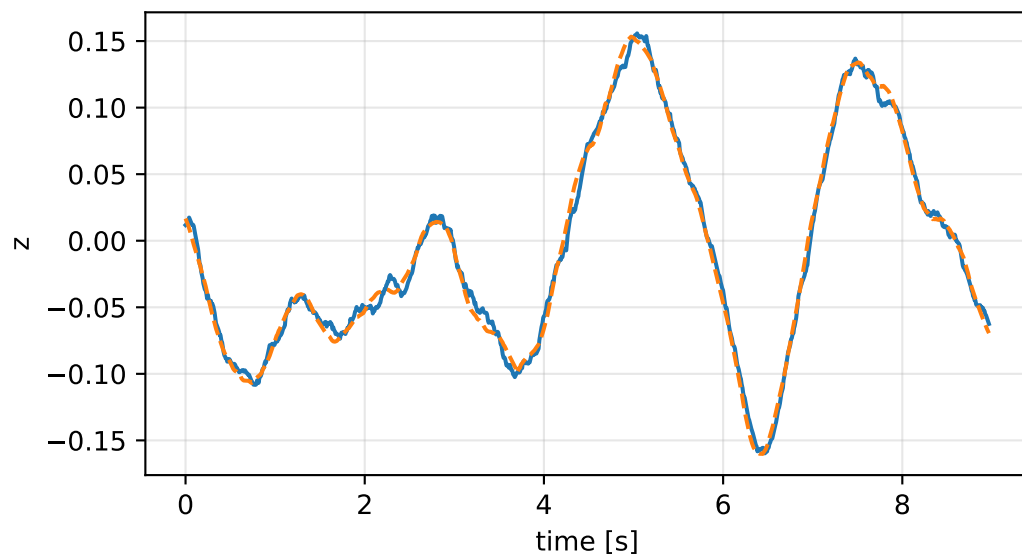
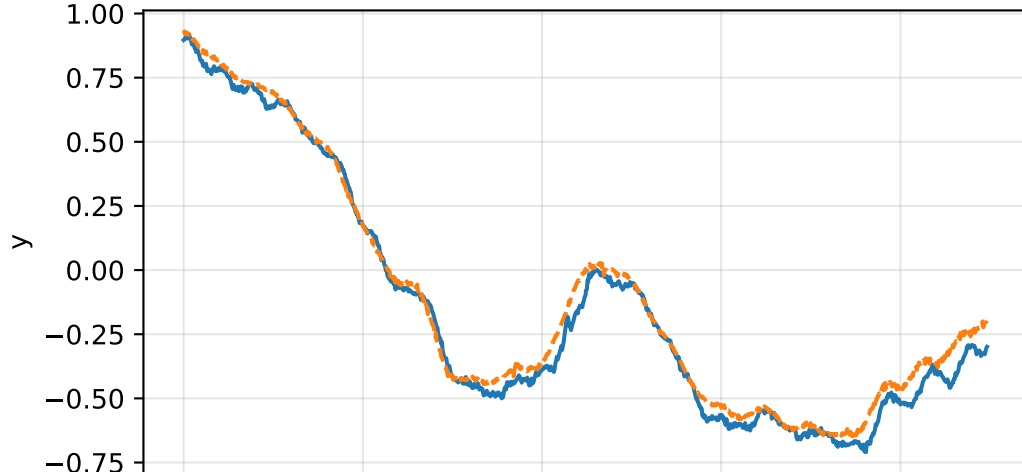
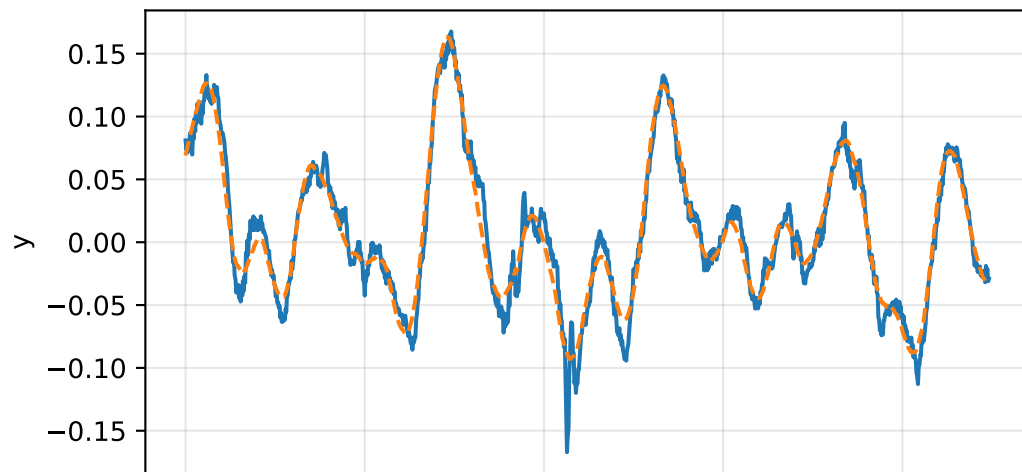
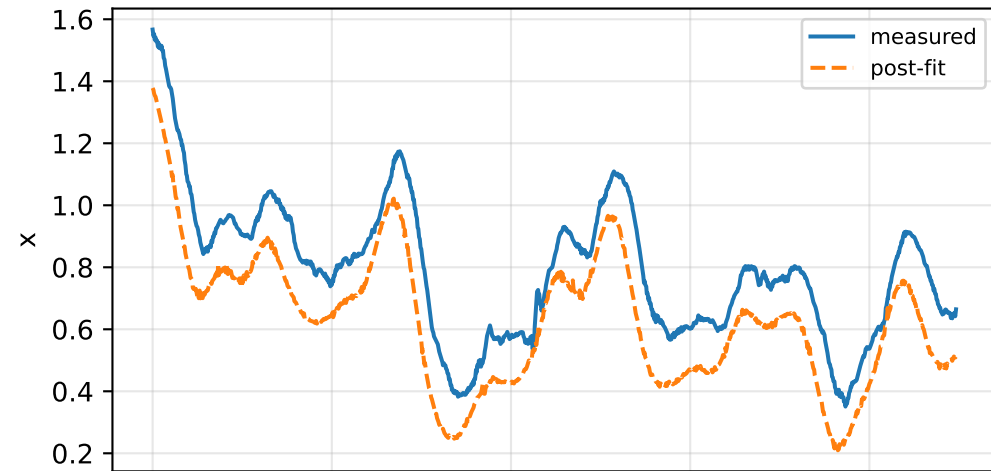


lag_fixed_one_period: IMU comparison (diagnostic only)

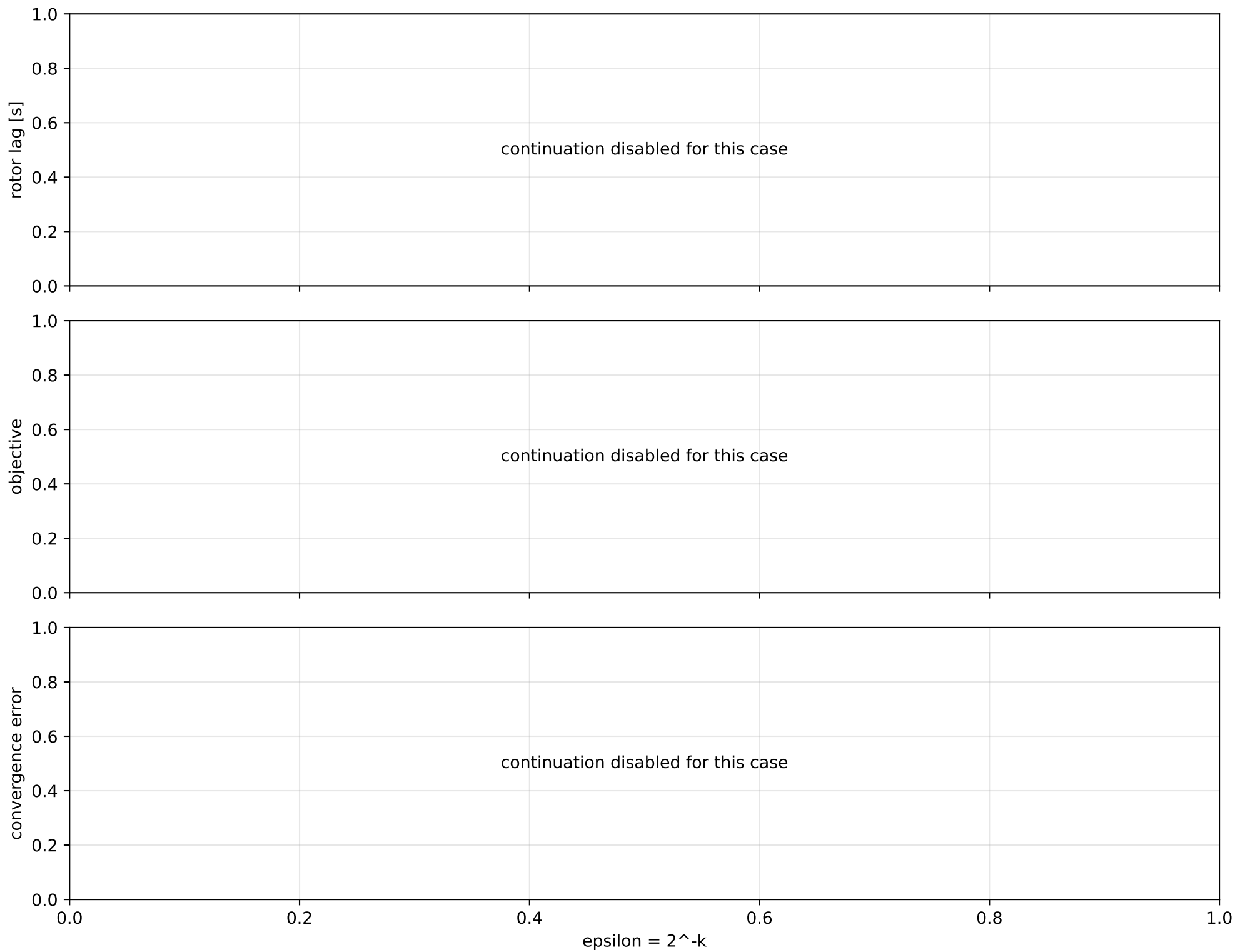
gyro [rad/s]



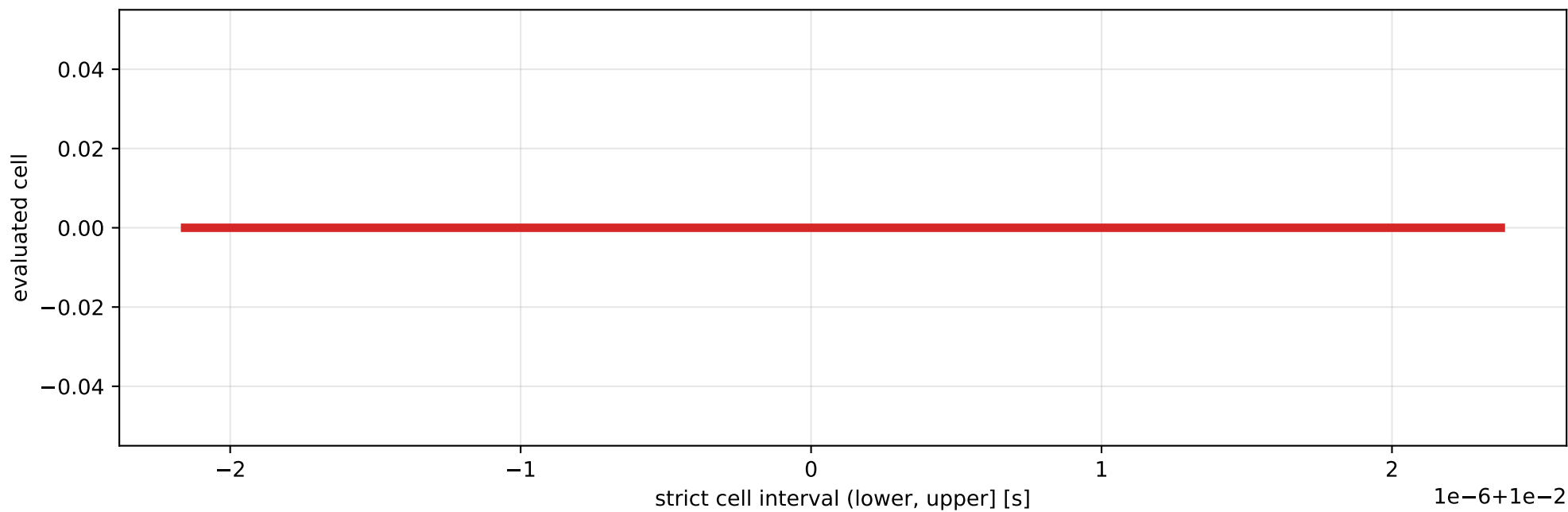
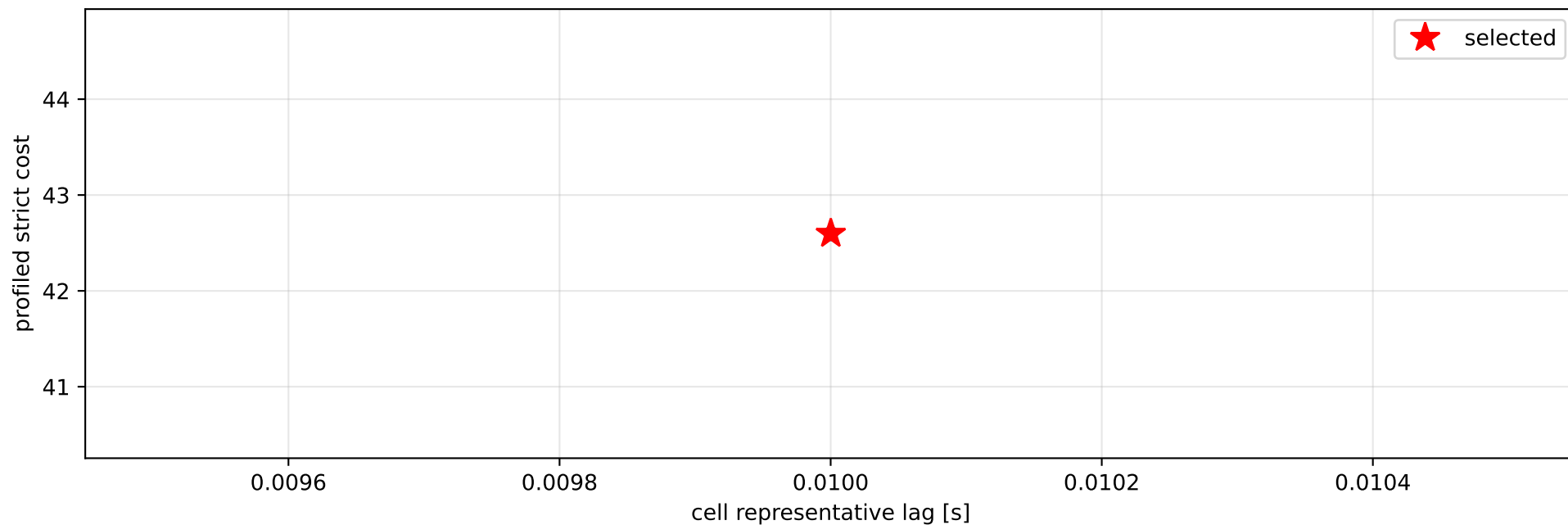
specific force [m/s^2]



lag_fixed_one_period: smooth-to-strict continuation



lag_fixed_one_period: exact strict-ZOH lag cells



selected (0.0099978447, 0.0100023746] s; final smooth 0.0100001097 s; data boundary=False

lag_fixed_one_period: parameter estimate

Case: lag_fixed_one_period

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.80221548

inertia [kg m²]:

```
[[ 3.41841615  0.05969596  0.18585702]
 [ 0.05969596  1.17502575 -1.42440464]
 [ 0.18585702 -1.42440464  2.46996605]]
```

CoG body [m]: [-0.09853535 -0.05726995 -0.14015861]

force effectiveness: [0.99604505 0.57164688 0.22452902 0.67183646]

scale-free J/m [m²]:

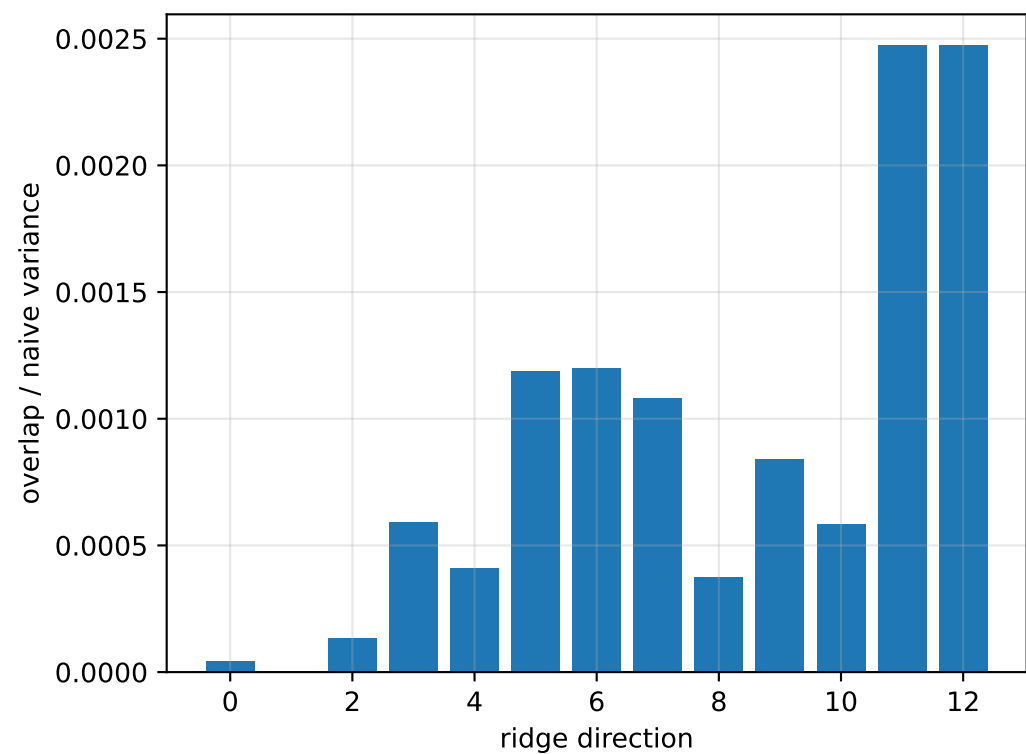
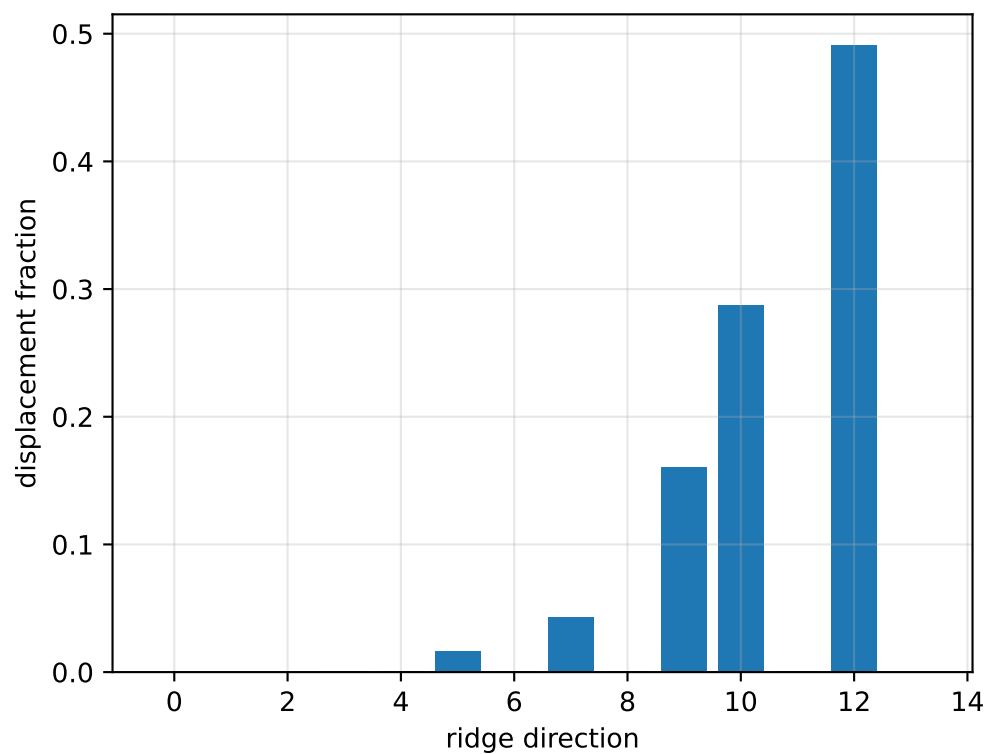
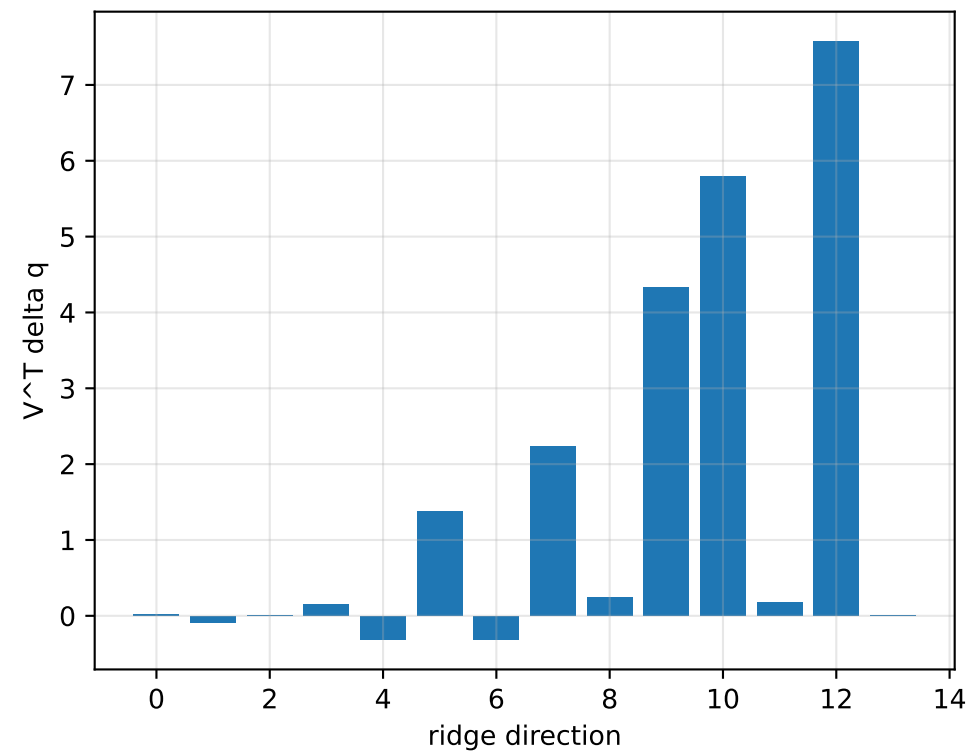
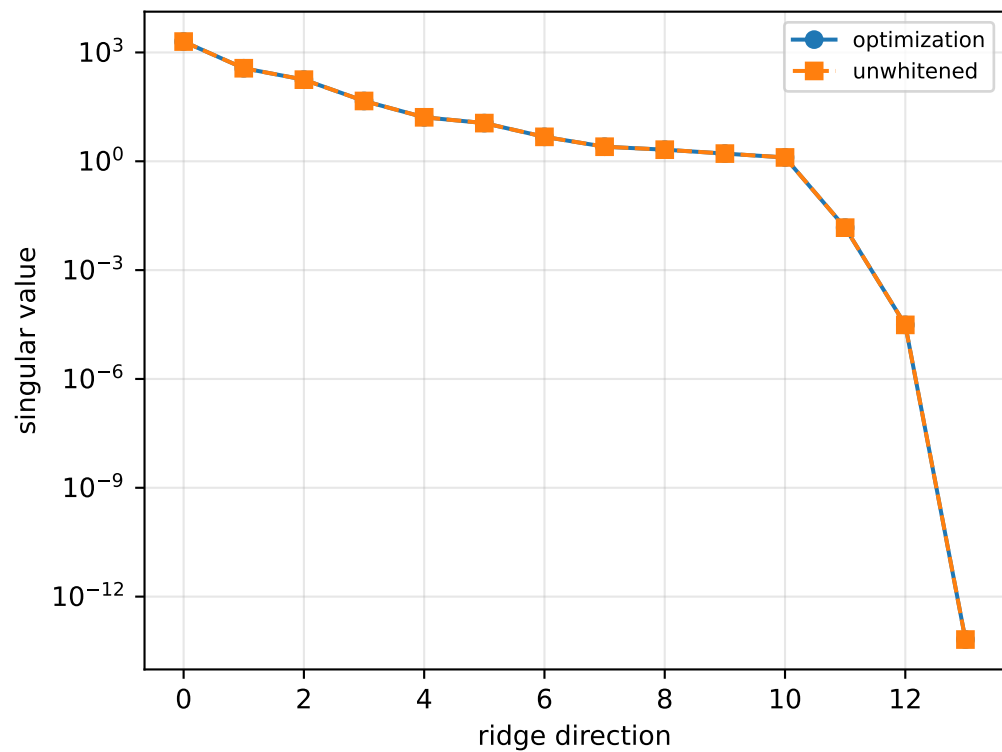
```
[[ 1.89678548  0.03312366  0.10312697]
 [ 0.03312366  0.6519896  -0.79036312]
 [ 0.10312697 -0.79036312  1.3705165  ]]
```

scale-free f/m: [0.55267812 0.3171912 0.12458501 0.37278365]

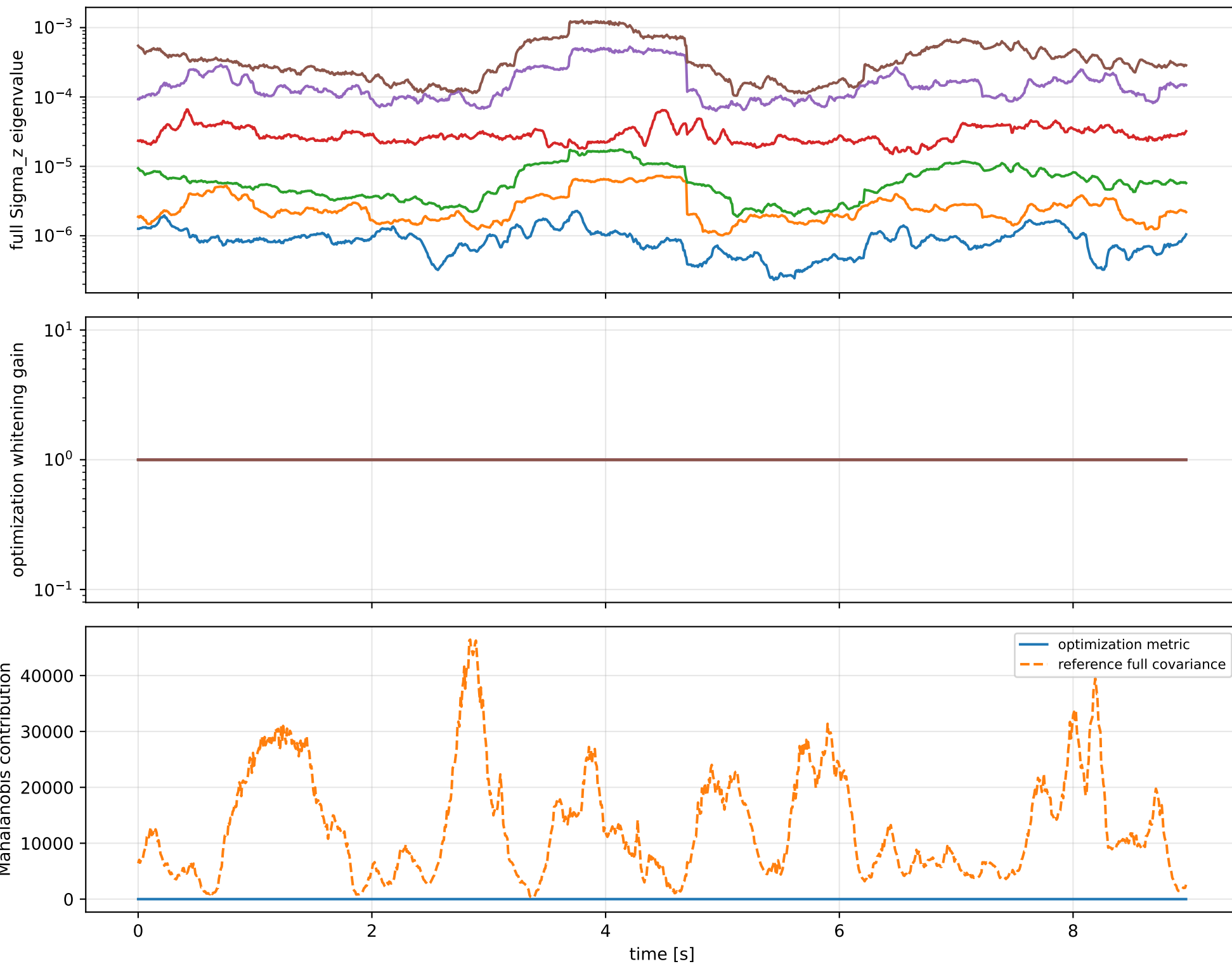
rotor lag cell: (0.0099978447, 0.0100023746] s

representative: 0.0100001097 s

lag_fixed_one_period: ridge spectrum (rank 13, nullity 1, ||Jv||=5.63e-14)



lag_fixed_one_period: optimization vs reference covariance



lag_fixed_one_period: wrench / closure (force max $5.5\text{e-}15$, torque max $3.33\text{e-}16$)

